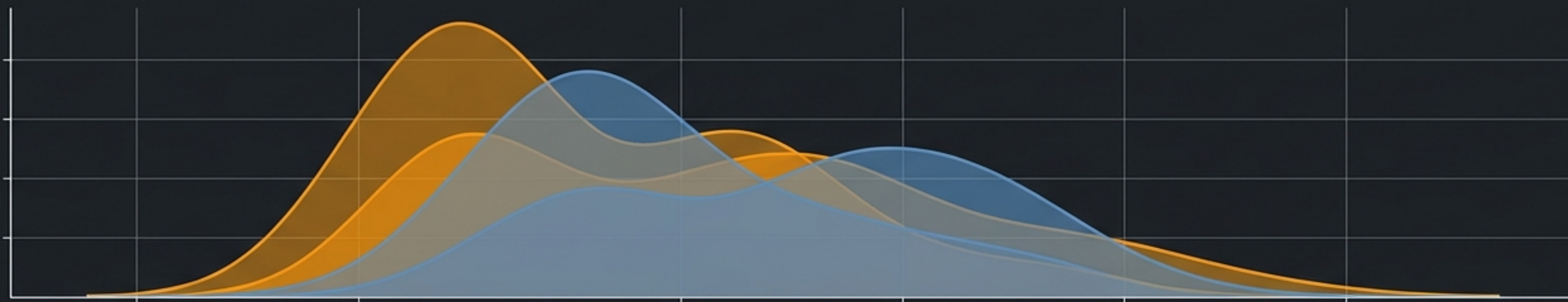


Shaping Data with Seaborn KDE Plots



A visual guide to understanding and coding Kernel Density Estimates—transforming scattered data points into smooth, continuous distributions.

Main Seaborn function for KDE visualization.

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Load dataset
df = sns.load_dataset('tips')
```

Defines variable for distribution and grouping by category.

```
# Create KDE Plot
```

```
sns.kdeplot(data=df, x='total_bill', hue='time', palette=['#FFA500', '#4682B4'], fill=True, common_norm=False, alpha=0.6)
plt.show()
```

Enables filled area and controls transparency for overlapping curves.

Specifies custom colors for different groups (editorial orange, soft blue).

Beyond Rigid Bins

Histogram



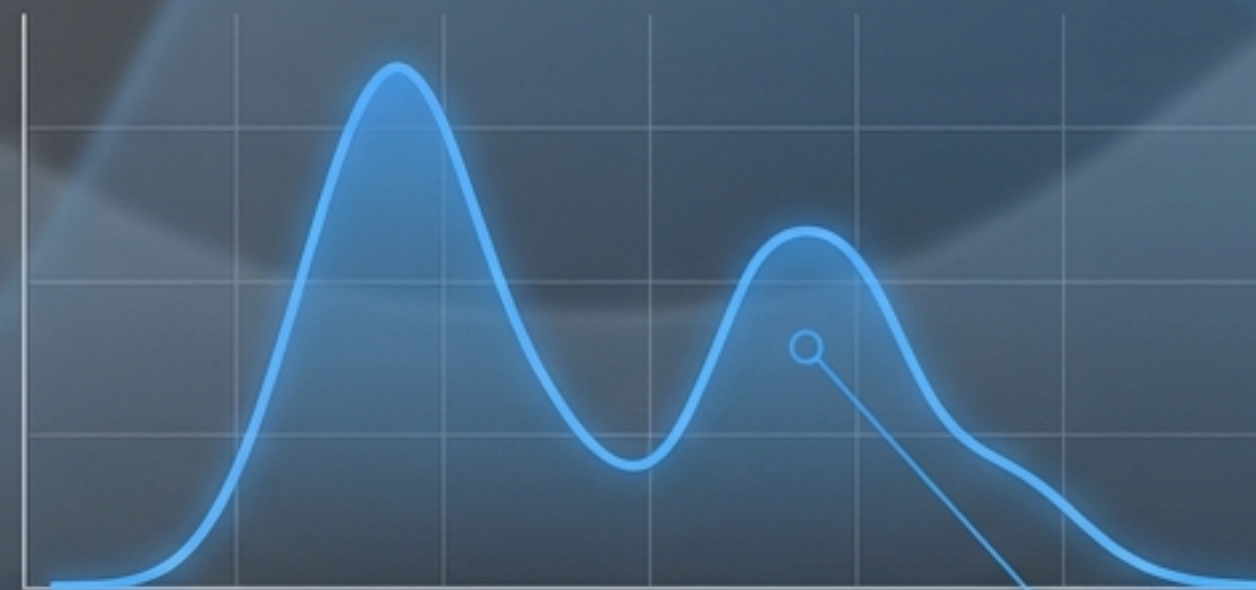
Rigid

Blocky Bins

Discrete

Forces data into fixed, rigid bins, obscuring the exact areas of concentration.

KDE Plot



Smooth

Flowing Curves

Continuous

Creates a continuous curve showing exact data data concentration. Peaks indicate high concentrations of data, while valleys indicate sparse data.

The Summation Engine Under the Hood

1. The Data Points



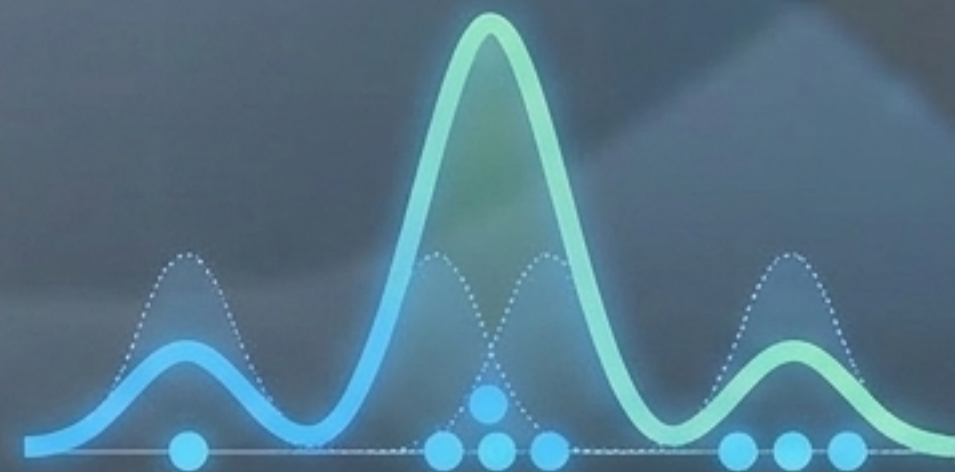
Imagine dropping all your data points onto a flat line.

2. The Kernels



Over every single point, the algorithm places a mathematical curve representing its density.

3. The Summation



The algorithm adds all those little curves together. Closely clustered points stack to create a giant peak.

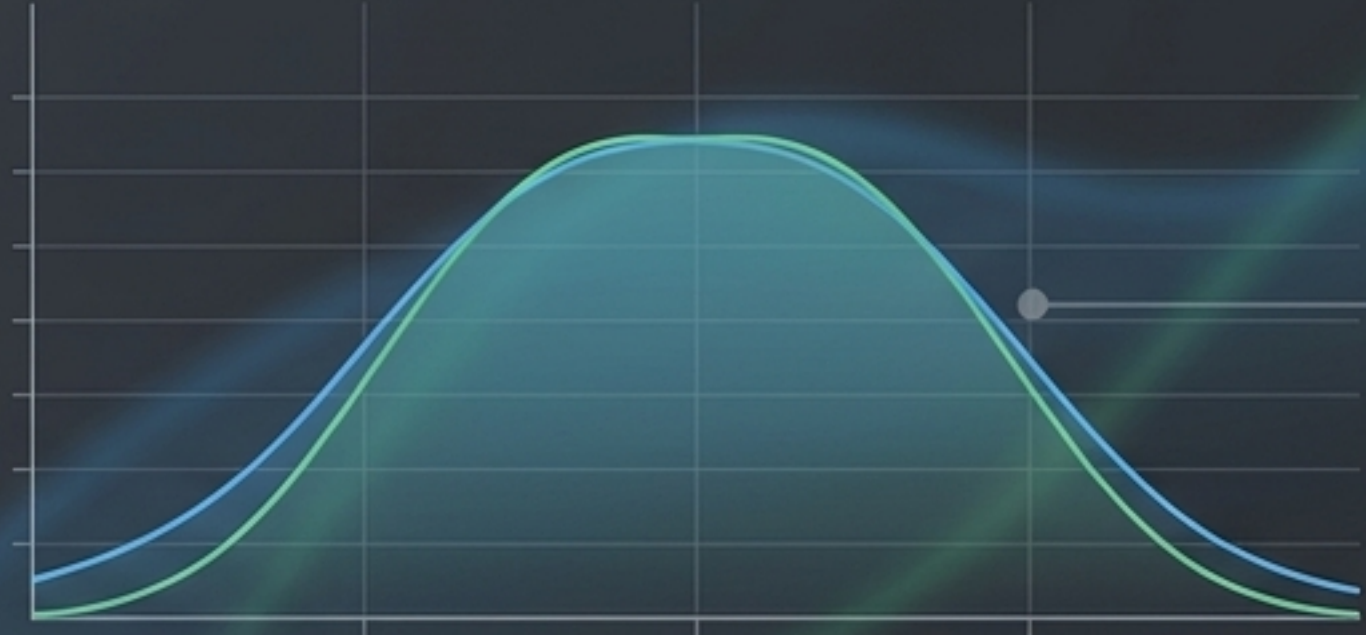
The Magic Parameter: Bandwidth

Low Bandwidth



- Kernels: Very thin and spiky.
- Result: Incredibly jagged and noisy.
- Effect: Tries to fit every tiny detail.

High Bandwidth



- Kernels: Very wide and fat.
- Result: Super smooth plot.
- Effect: Might lose important details and hide multiple peaks.

From Theory to Syntax

The dataset holding your variables.

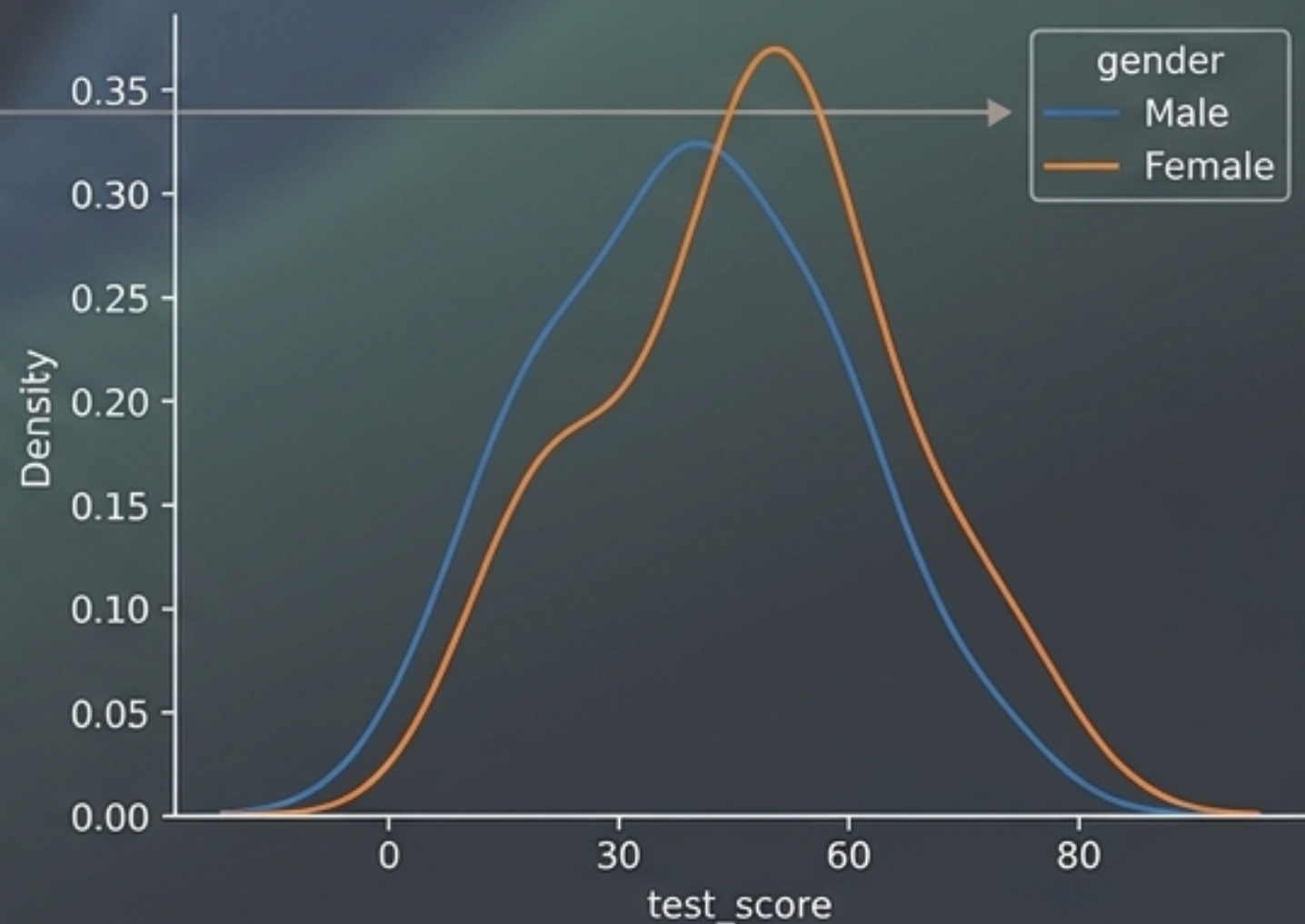
```
sns.kdeplot(data=..., x=..., hue=...)
```

The category to split the data into overlapping, distinct curves.

The specific column determining the X-axis distribution.

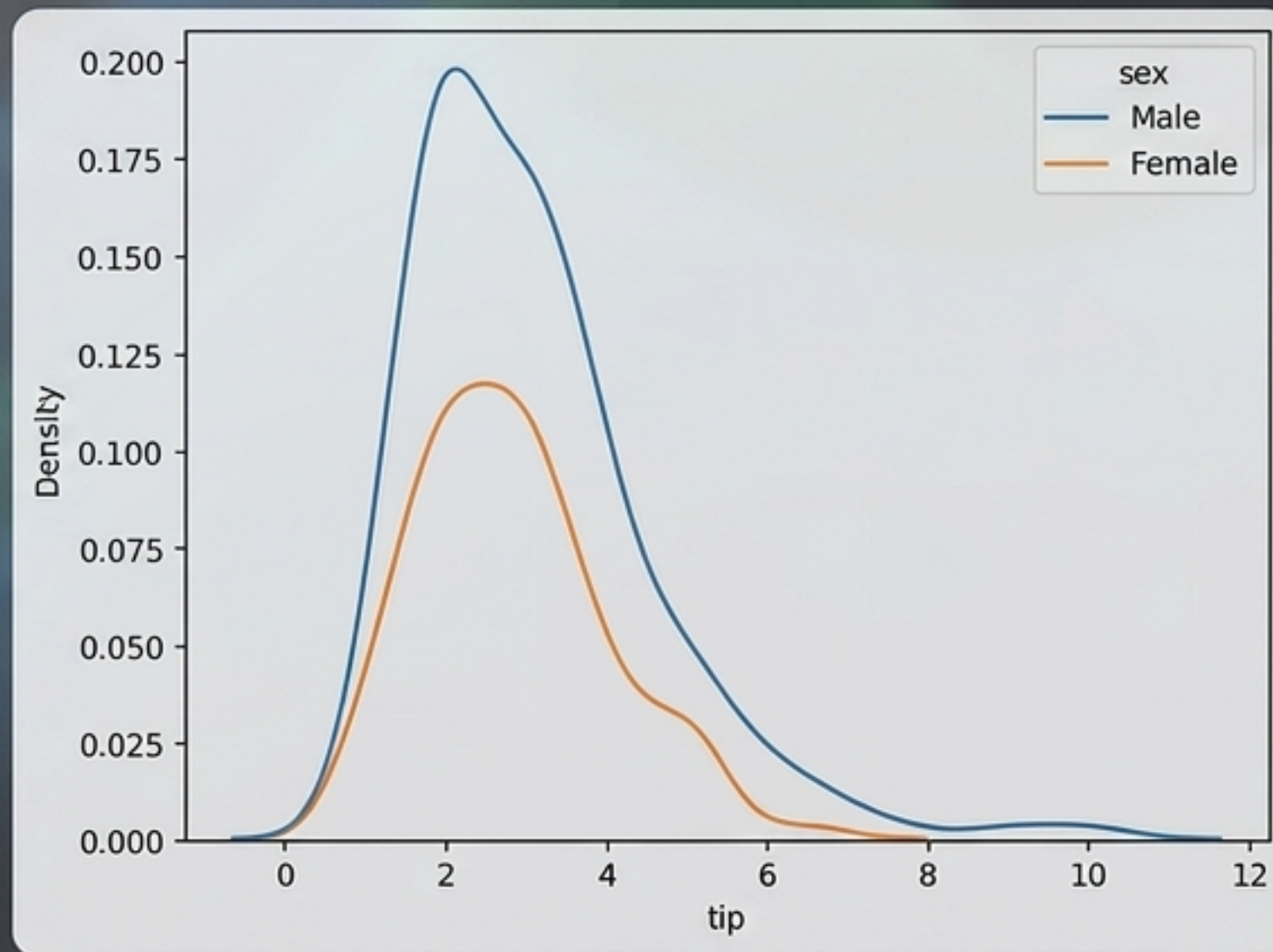
Mapping Student Test Scores

```
# figure level  
sns.kdeplot(data=student,  
            x="test_score", hue='gender')
```



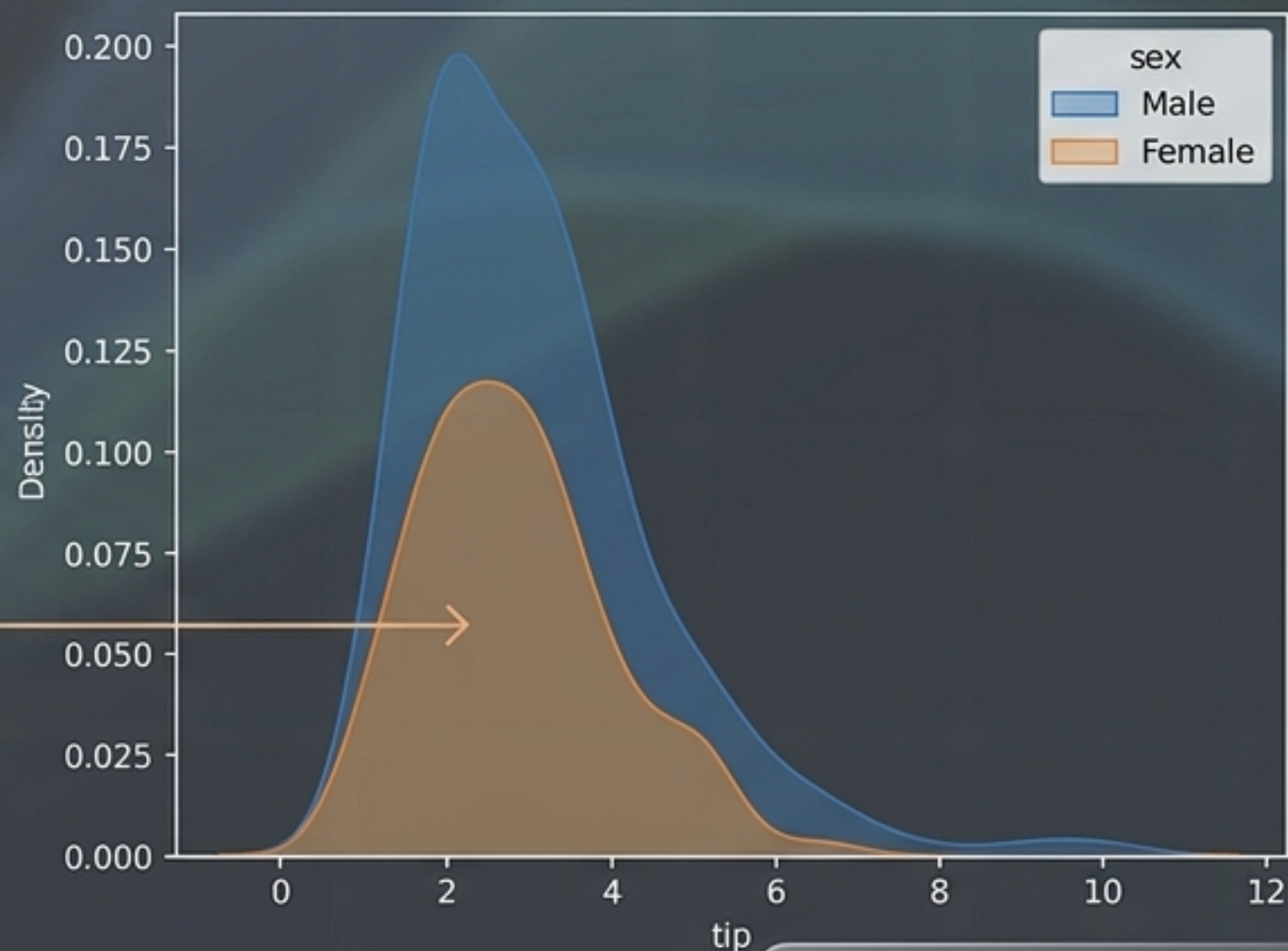
Analyzing Restaurant Tips

```
sns.kdeplot(data=tips_data,  
            x="tip",  
            hue="sex")
```



Enhancing Readability with Fill

```
sns.kdeplot(data=tips_data,  
            x="tip",  
            hue="sex",  
            fill=True)
```



Adding fill improves contrast when overlapping distributions.

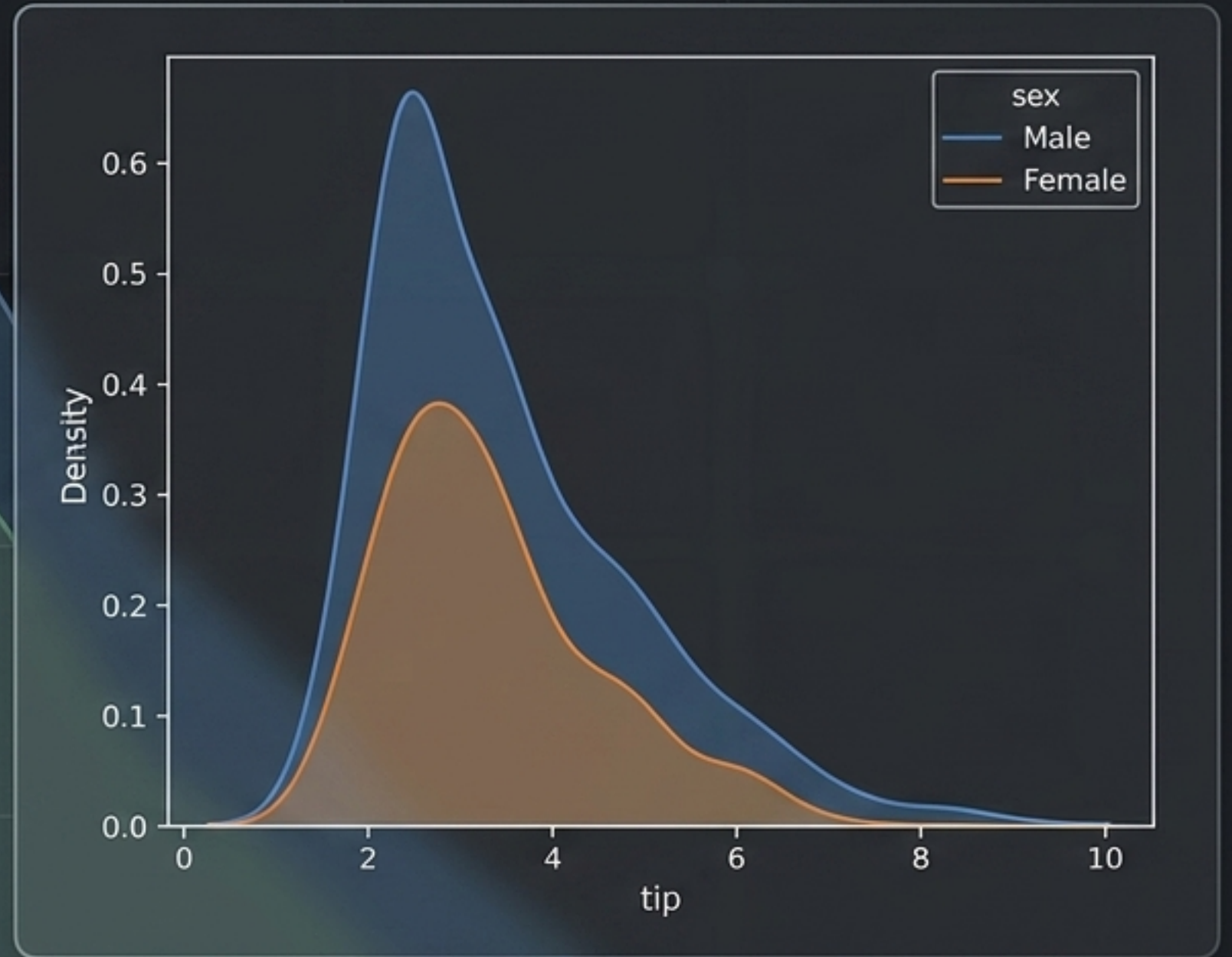
Fine-Tuning the Smoothing Engine

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.kdeplot(data=tips_data, x="tip",
            hue="sex", bw_adjust=1.0, fill=True)

plt.show()
```

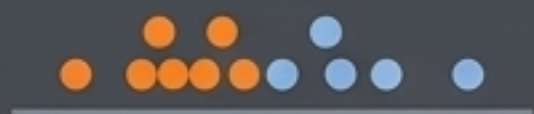
This scales the theoretical bandwidth. Lower values create spiky curves; higher values smooth out the peaks.



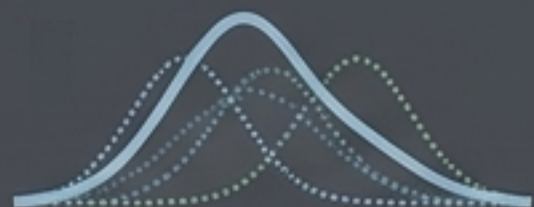
Code-to-Concept Translation

The Theory

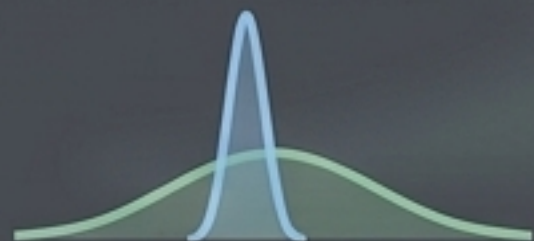
The Data Points



The Summation



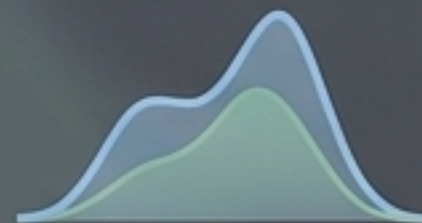
Kernel Width



The Code Parameter

```
data=tips_data, x="tip"
```

The resulting
`sns.kdeplot()` curve



```
bw_adjust=  
(The precise lever controlling smoothness)
```